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Evaluation of Antidiarrheal Potential of *Emblica officinalis*

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Abstract

The antidiarrheal potential of the methanol extract of the fruit of *Emblica officinalis* Gaertn. (Euphorbiaceae) was evaluated using several experimental models of diarrhea in Wistar albino rats. The methanol extract showed a significant inhibitory effect on rats with diarrhea induced by castor oil and magnesium sulfate. The methanol extract produced a significant reduction in gastrointestinal motility in charcoal meal tests in rats. It also significantly inhibited PGE₂-induced enteropooling as compared to control animals. The results obtained establish the efficacy and substantiate the use of this herbal remedy as a nonspecific treatment for diarrhea in folk medicine.

Keywords: Antidiarrheal, *Emblica officinalis*, Euphorbiaceae, gastrointestinal motility, intestinal secretion.

Introduction

Emblica officinalis Gaertn. (Syn: *Phyllanthus emblica*) is a member of a small genus *Emblica* (Euphorbiaceae). It grows in tropical and subtropical parts of China, India, Indonesia, and the Malay Peninsula. The fruit is commonly known as amla or emblic myrobalan and is highly valued in traditional Indian medicine. In Unani, the dried fruits are used to treat hemorrhage, diarrhea, and dysentery. A sherbet made of the fruits and lemon juice reportedly arrests acute bacillary dysentery (Parrotta, 2001).

Several constituents of *E. officinalis* fruits have been identified, mainly the hydrolyzable tannins (10% to 12% in pericarp), emblicanin A, emblicanin B, punigluconin, and pedunculagin. Emblicanin A and B have

been proposed to be the active constituents with significant *in vitro* antioxidant activity (Ghosal et al., 1996). It also contained gallic and ellagic acids that inhibit the degradation of vitamin C (Damodaran & Nair, 1936). Earlier studies have demonstrated potent antimicrobial (Ahmad et al., 1998), antioxidant (Bhattacharya et al., 1999), adaptogenic (Rege et al., 1999), hepatoprotective (Jeena et al., 1999), antitumor (Jose et al., 2001), and antiulcerogenic (Sairam et al., 2002) activities in the fruits of *E. officinalis*. In previous studies, we have reported the anti-inflammatory, analgesic, and antipyretic activity of *E. officinalis* fruits (Sharma et al., 2003; Perianayagam et al., 2004). In this investigation, we report the antidiarrheal effect of the methanol extract of the fruit of *E. officinalis* using several experimental models of diarrhea in rats.

Materials and Methods

Plant material

The fruits of *E. officinalis* were collected during the month of November 2000 from Vadalur, Tamilnadu, southern India. The plant material was taxonomically identified and authenticated by Prof. P. Jayaraman, Taxonomist, Plant Anatomy Research Centre, Chennai, India. A voucher specimen has been deposited in the Herbarium of the Department of Pharmacognosy, K.P. College of Pharmacy, Tiruvannamalai, Tamilnadu, India, for future reference. The pulp of fresh fruit was separated from the hard seed by pounding in a stone mortar, dried rapidly at 60°C, and the dry material

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reduced to small pieces, pulverized using a mechanical grinder, and passed through 40 mesh and stored in an air-tight container for further use.

Preparation of extract

The powdered fruits (1 kg) were extracted successively with petroleum ether, chloroform, and methanol at room temperature. After exhaustive extraction, the methanol extract (MEO) was concentrated under reduced pressure to a syrupy consistency. A light brown concentrate was obtained (yield 6.2% w/w with respect to the dry starting material) and kept in a desiccator for further use. The chemical constituents of the extract were identified by qualitative chemical tests and further confirmed by thin-layer chromatography for the presence of phenolic compounds and tannins. The methanol extract was suspended in carboxy methylcellulose solution (1%) in the desired concentration just before use.

Animals used

Albino (Wistar) rats (200–230 g) of either sex were maintained at uniform laboratory conditions in standard polypropylene cages and provided with food and water *ad libitum*. The animals were acclimatized for a period of 14 days prior to performing the experiments.

Castor oil-induced diarrhea in rats

The method reported by Awouters et al. (1978) with modifications was used in the current study. Rats of either sex (200–230 g) were fasted for 18 h; they were then divided into five groups of five individuals. The MEO extract was administered orally at doses of 50, 100, and 150 mg/kg by gavage as suspension to the first three groups of animals. The fourth group received loperamide (3 mg/kg) orally as suspension (positive control). The fifth group, which served as the control, was treated with carboxy methylcellulose suspension. After 60 min of treatment, the animals of each group received 1 ml of castor oil orally, by gavage, and the consistency of fecal material and the frequency of defecation was noted up to 4 h in the transparent plastic dishes placed beneath the individual rat cages (Gnansekar & Perianayagam, 2004).

Magnesium sulfate-induced diarrhea in rats

The method reported by Doherty (1981) was used in the current study. Rats of either sex (200–230 g) were fasted for 18 h; they were then divided into five groups of five individuals. The first three groups of animals were treated orally with 50, 100, and 150 mg/kg of MEO

extract. The fourth group received loperamide (3 mg/kg) orally as suspension (positive control). The fifth group, which served as the control, was treated with aqueous carboxy methylcellulose suspension. After 60 min of treatment, the animals of each group were treated with magnesium sulfate at a dose of 2 g/kg and the consistency of fecal material and the frequency of defecation was noted up to 4 h in the transparent plastic dishes placed beneath the individual rat cages (Gnansekar & Perianayagam, 2004).

Gastrointestinal motility tests

Rats were fasted for 18 h and then placed in five cages containing five individuals in each cage. Each animal was treated orally with 1 ml of charcoal meal [5% deactivated charcoal (5%) in aqueous tragacanth (10%)], followed by oral administration of MEO suspension to three groups of animals in doses of 50, 100, and 150 mg/kg. The fourth group received atropine (0.1 mg/kg, i.p.); which is the standard drug for comparison and the fifth group was treated with aqueous carboxy methylcellulose suspension (vehicle control). After 30 min, each animal was sacrificed and the intestinal distance moved by the charcoal meal from the pylorus was cut, measured, and expressed as a percentage of the distance from the pylorus to cecum for each animal (Gunakunru et al., 2005).

Posterglandin E₂ (PGE₂)-induced enteropooling

In this method, rats were deprived of food and water for 18 h and placed in five cages, with five animals per cage. The first three groups were treated with 50, 100, and 150 mg/kg doses of MEO. The fourth group was treated with 1 ml of ethanol (5% v/v) in normal saline (i.p.) and then it was treated with aqueous carboxy methylcellulose suspension, which served as vehicle control. Immediately after the extract administration, PGE₂ was administered orally to each rat (100 µg/kg) in the first three groups. The fifth group was treated with PGE₂ (100 µg/kg) as well as aqueous acacia suspension and served as the PGE₂ control group. Thirty min after administration of PGE₂, each rat was sacrificed, and the whole length of the intestine from the pylorus to the cecum was dissected out, its content collected in a test tube, and the volume measured (Gunakunru et al., 2005).

Statistical analysis

All data are expressed as the mean $\pm$ SEM. The difference in response to test drugs and control was evaluated using one-way analysis of variance followed by Dunnett's *t*-test. *p* values less than 0.05 imply significance.

Table 1. Effect of MEO on castor oil-induced diarrhea in rats.

Oral pretreatment at 0 h + castor oil at 1 h	Mean number of wet feces in 4 h
Control	8.2 ± 1.35
LPA (3 mg/kg)	0.00*
MEO (50 mg/kg)	3.4 ± 1.20*
MEO (100 mg/kg)	2.2 ± 1.01*
MEO (150 mg/kg)	1.4 ± 0.73*

Results are mean ± SEM, n = 5; *p < 0.01 compared with control, Dunnett's *t*-test after analysis of variance. MEO, methanol extract of *E. officinalis*; LPA, loperamide.

Results

Chemical analysis

The presence of phenolic compounds and tannins were detected on preliminary phytochemical screening of the dried extract.

Inhibition of castor oil-induced diarrhea

Administration of castor oil produced characteristic semisolid diarrhea droppings in 18-h starved rats of the control groups during the 4-h observation period. When the doses of MEO extract were administered orally 1 h before the administration of castor oil, the time in the excretion of the first diarrheal feces was significantly increased compared to the control group. It also significantly reduced the number of wet feces excreted in the 4 h after administration of castor oil (Table 1). Treatment with the reference compound, loperamide (3 mg/kg), also exhibited significant antidiarrheal activity in castor oil-induced diarrhea.

Inhibition of magnesium sulfate-induced diarrhea

Administration of magnesium sulfate caused characteristic diarrhea droppings, which was maximal at 3 h after oral administration in rats. MEO extract significantly

Table 2. Effect of MEO on magnesium sulfate-induced diarrhea in rats.

Oral pretreatment at 0 h + MgSO ₄ at 1 h	Mean number of wet feces in 4 h
Control	6.2 ± 1.15
LPA (3 mg/kg)	0.00*
MEO (50 mg/kg)	2.6 ± 0.81*
MEO (100 mg/kg)	1.0 ± 0.58*
MEO (150 mg/kg)	0.60 ± 0.40*

Results are mean ± SEM, n = 5; *p < 0.01 compared with control, Dunnett's *t*-test after analysis of variance. MEO, methanol extract of *E. officinalis*; LPA, loperamide.

Table 3. Effect of MEO extract on small intestinal transit.

Oral pretreatment	% Movement of charcoal meal
Control	83.1 ± 3.30
ATP (0.1 mg/kg)	38.3 ± 2.12*
MEO (50 mg/kg)	54.4 ± 3.99*
MEO (100 mg/kg)	43.8 ± 2.64*
MEO (150 mg/kg)	40.8 ± 1.88*

Results are mean ± SEM, n = 5; *p < 0.01 compared with control, Dunnett's *t*-test after analysis of variance. MEO, methanol extract of *E. officinalis*; ATP, atropine.

reduced the excretion of diarrheal stools during the 4-h period as compared to control animals (Table 2). Loperamide (3 mg/kg) also caused significant reduction in the number of wet fecal droppings in magnesium sulfate-induced diarrhea.

Effect on small intestinal transit

The MEO extract at doses of 50, 100, and 150 mg/kg decreased the propulsion of charcoal meal through the gastrointestinal tract, as compared with the control group. Atropine (0.1 mg/kg) reduced the motility of the intestine to a greater extent (p < 0.001) (Table 3).

Antienterpooling activity

The extract (MEO) at doses of 50, 100, and 150 mg/kg significantly inhibited PGE₂-induced enterpooling in rats (Table 4). PGE₂ induced a significant increase in the fluid volume of the rat intestine when compared with vehicle control group; which received ethenol in normal saline.

Discussion

In the current study, the methanol extract of *E. officinalis* exhibited dose-related antidiarrheal activity at doses of

Table 4. Antienterpooling effect of MEO extract in rats.

Treatment	Volume of intestinal fluid (ml)	p values
Ethanol in saline	0.86 ± 0.12	
PGE ₂ in ethanol (100 µg/kg)	2.72 ± 0.27	0.001 ^a
MEO (50 mg/kg)	1.74 ± 0.20	0.05 ^b
MEO (100 mg/kg)	1.32 ± 0.24	0.01 ^b
MEO (150 mg/kg)	1.10 ± 0.15	0.01 ^b

Results are mean ± SEM, n = 5, ANOVA test followed by Dunnett's *t*-test. MEO, methanol extract of *E. officinalis*.

^aWith respect to ethanol in saline treatment.

^bWith respect to PGE₂ treatment.

50, 100, and 150 mg/kg on diarrhea induced by castor oil or magnesium sulfate. The extract had a similar activity as loperamide when tested at doses of 50, 100, and 150 mg/kg and significantly inhibited the frequency of defecation and the wetness of the fecal droppings when compared to control rats.

Castor oil or its active component, ricinolic acid, induces permeability changes in mucosal fluid and electrolyte transport that results in a hypersecretory response and diarrhea (Ammon et al., 1974; Gaginella et al., 1975). The experimental studies in rats demonstrated a significant increase in the portal venous PGE₂ concentration after oral administration of castor oil (Luderer et al., 1980). Ricinolic acid markedly increased the PGE₂ content in the gut lumen and also caused an increase of the net secretion of the water and electrolytes into the small intestine (Beubler & Juan, 1979). Delay in castor oil-induced diarrhea is a characteristic feature of NSAIDs (Awouters et al., 1978). Like NSAIDs, *E. officinalis* has also been shown to possess potent anti-inflammatory, analgesic, and antipyretic activity (Sharma et al., 2003; Perianayagam et al., 2004). Thus, the antidiarrheal activity of the methanol extract of *E. officinalis* may be due to inhibition of prostaglandin synthesis and release.

The methanol extract also inhibited magnesium sulfate-induced diarrhea, which is presumed to act by its osmotic properties and by cholecystokinin production (Melo et al., 1988). Therefore, mechanisms other than inhibition of prostaglandin synthesis must also be considered as a number of factors that contribute to the functions of the gut. Examples of these include cholinergic, adrenergic, tryptaminergic, opiate system, α_2 -adrenoceptor stimulants, and vasoactive intestinal polypeptides.

The extract appears to act on all parts of the intestine. Thus, it reduced the intestinal propulsive movement in the charcoal meal-treated model; at 150 mg/kg, MEO showed activity similar to that of atropine. These observations tend to suggest that the MEO extract at doses of 50, 100, and 150 mg/kg reduced diarrhea by inhibiting gastrointestinal motility.

The extract (MEO) at different dose levels (50, 100, and 150 mg/kg) significantly inhibited PGE₂-induced intestinal fluid accumulation (enteropooling). It has been shown that E type prostaglandins cause diarrhea in experimental animals as well as human beings (Eakins & Sanner, 1972). Their mechanism has been associated with dual effects on gastrointestinal motility as well as on water and electrolyte transport (Dajani et al., 1975). Based on these observations, it seems reasonable to suggest that the antidiarrheal effect of methanol extract may be due to the inhibition of prostaglandin actions.

Earlier reports of phytochemical studies of *E. officinalis* fruit indicate the presence of hydrolyzable tannins (10% to 12% in pericarp), embicanin A, emblicanin B, punigluconin, and pedunculagin as major components.

Emblicanin A and B have been proposed to be the active constituents with significant *in vitro* antioxidant activities (Ghosal et al., 1996). Fruit of *E. officinalis* also contains ellagotannin (Damodaran & Nair, 1936), trigallayl glucose (Theresa, 1968), phyllembin (Rao & Siddiqui, 1964), phyllemblic acid, and emblicol (Pillay & Mahadeva Iyer, 1958). Previous studies have demonstrated the antidiarrheal activity of tannin containing plant extracts (Mukherjee et al., 1998; Rani et al., 1999; Sairam et al., 2003). The antidiarrheal activity of tannins has been ascribed to denaturing proteins forming protein tannates, which makes the intestinal mucosa more resistant and reduces secretion (Tripathi, 1994). The phytochemical analysis of the extract using different methods indicated the presence of phenolic compounds and tannins, which might be in part responsible for the antidiarrheal activity of methanol extract of *E. officinalis*.

The current results indicate that the methanol extract of *Embolica officinalis* possesses significant antidiarrheal activity due to its inhibitory effect on both gastrointestinal propulsion and fluid secretion. The inhibitory effect of the extract justifies the use of the plant as a nonspecific antidiarrheal agent in folk medicine.

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